

HEART ATTACK PREDICTION USING DEEP LEARNING APPROACH BASED ON ECG SIGNAL



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ABSTRACT

A CNN-based model designed to detect myocardial infarction from ECG images.

Through advanced preprocessing and data augmentation, it achieves high accuracy and robustness.

The system enables fast, precise, and non-invasive early detection to support clinical diagnosis.

INTRODUCTION

OBJECTIVE

To detect myocardial infarction (heart attack) from ECG images using deep learning, enabling early, accurate, and non-invasive diagnosis.

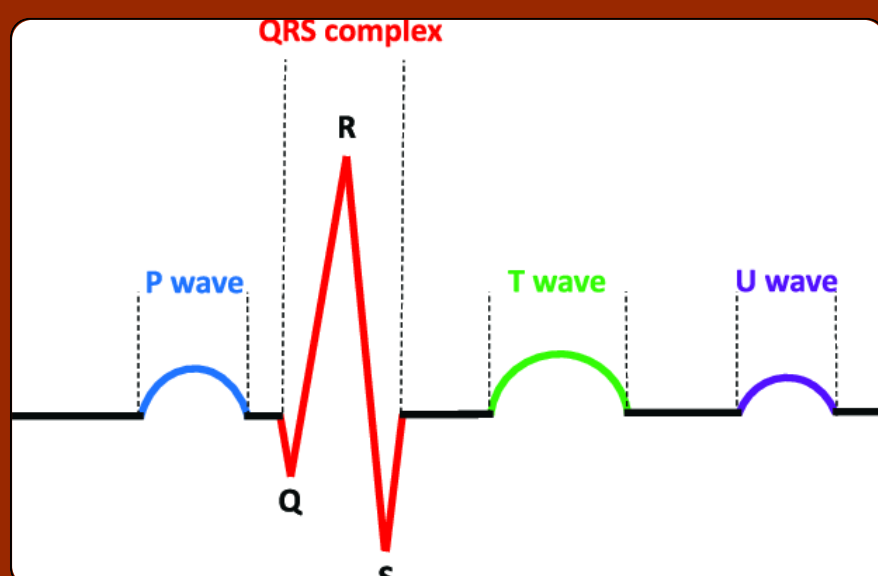
TECHNOLOGY

- Convolutional Neural Network (CNN) model for ECG image classification
- Preprocessing and data augmentation for improved accuracy and robustness
- Automated detection system supporting real-time clinical decision-making

OUTCOME

- High accuracy in identifying heart attack patterns
- Fast and reliable diagnostic support
- Reduces manual interpretation time for healthcare professionals

ECG WAVEFORM EXPLANATION



- P wave: Atrial depolarization (atria contraction)
- QRS complex: Ventricular depolarization (main diagnostic feature)
- T wave: Ventricular repolarization (recovery phase)
- U wave: Repolarization of Purkinje fibers (may not always appear)

Abnormalities in the QRS complex, ST segment, or T wave often indicate a heart attack.

REFERENCE

- S. B. Patil and D. Kumaraswamy, "Extraction of significant patterns from heart disease warehouses for heart attack prediction," *IJCSNS International Journal of Computer Science and Network Security*, vol. 9, no. 2, pp. 228–235, 2009.
- S. Nashif, M. R. Raihan, M. R. Islam, and M. H. Imam, "Heart disease detection by using machine learning algorithms and a real-time cardiovascular health monitoring system." *World Journal of Engineering and Technology*, vol. 6, no. 4, pp. 854–873, 2018.

PROPOSED ALGORITHM

ALGORITHM WORKFLOW

1. DATA COLLECTION & PREPROCESSING

- Gather ECG images from trusted medical datasets.
- Normalize, resize, and filter noise for consistent input quality.

2. Image Segmentation & Augmentation

- Focus on key ECG features (P wave, QRS complex, T wave).
- Apply rotation, flipping, and scaling to enhance dataset diversity and robustness.

3. Model Selection, Training & Evaluation

- Implement a Convolutional Neural Network (CNN) model.
- Train using labeled ECG images.
- Validate and evaluate model performance with metrics like accuracy, precision, and recall.

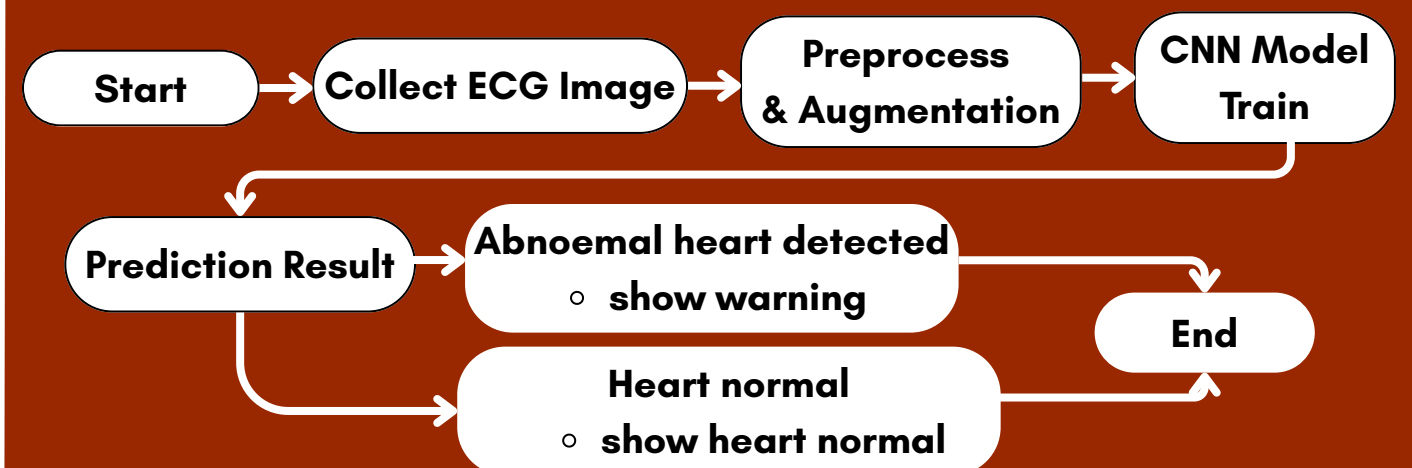
4. Prediction Phase

If Abnormal Condition Detected:

- Display "Warning: Abnormal Heart Detected. ❌"
- Store diagnostic results in the database.

If Normal Condition:

- Display "Heart Condition Normal. ✅"
- Continue monitoring or wait for the next ECG input.

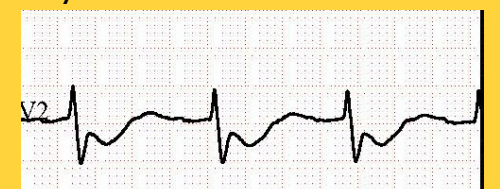


METHODOLOGY

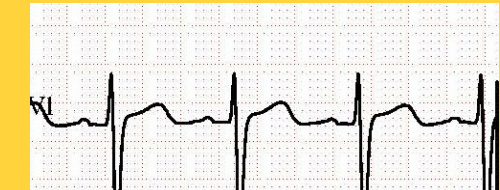
Normal ECG



Myocardial Infarction



Abnormal Heartbeat



The proposed system employs machine learning to detect heart conditions from ECG images. A dataset containing Normal, Myocardial Infarction, and Abnormal Heartbeat ECG reports is utilized. The images undergo preprocessing steps such as resizing, denoising, and normalization before being fed into a CNN model for feature extraction and classification. The trained model effectively differentiates between the three cardiac conditions with high accuracy.

CONCLUSION

In conclusion, this project effectively uses deep learning and image processing to predict heart attack risk from ECG data. With preprocessing and augmentation, it delivers an accurate, automated tool to aid early diagnosis and reduce mortality.